READ FIRST · UNPACK THE ABBREVIATIONS

CPU (Central Processing Unit)

OS (Operating System)

COMPUTER SCIENCE / DATA STRUCTURES / YEAR 1

# THE COMPUTER, EXPLAINED

A phone, laptop and desktop computer share the same basic idea.

SCAN & LEARN

## WHAT HAPPENS • FOLLOW THE ARROWS

01 YOU PRESS A BUTTON → 02 THE APP REQUESTS → 03 THE OS COORDINATES → 04 HARDWARE WORKS

## FOUR IDEAS TO KEEP

### 1 HARDWARE = physical parts

Things you can touch: screen, processor, memory, storage. The processor (CPU) executes instructions.

Picture a kitchen with its tools and appliances.

### 2 SOFTWARE = instructions

Apps are programs: browser, game, camera. Code tells them which operations to perform.

Picture the recipe used by a cook.

### 3 OS = the coordinator

Windows, Linux, macOS and Android are operating systems. They manage apps, files and devices.

The OS is NOT the browser. The browser is an app.

### 4 INPUT → PROCESS → OUTPUT

Input: click or key. Processing: instructions run. Output: an image, sound or saved file.

Example: press Volume Up → the sound changes.

## TRY IT — NO COMPUTER NEEDED

**One student is the user, one an app, one the OS, one the CPU. Act out pressing “Play”.**

## THE ONE SENTENCE TO REMEMBER

**The SCREEN shows the result; apps, the OS and hardware do the work.**

Question: Is Android hardware, an app or an operating system?